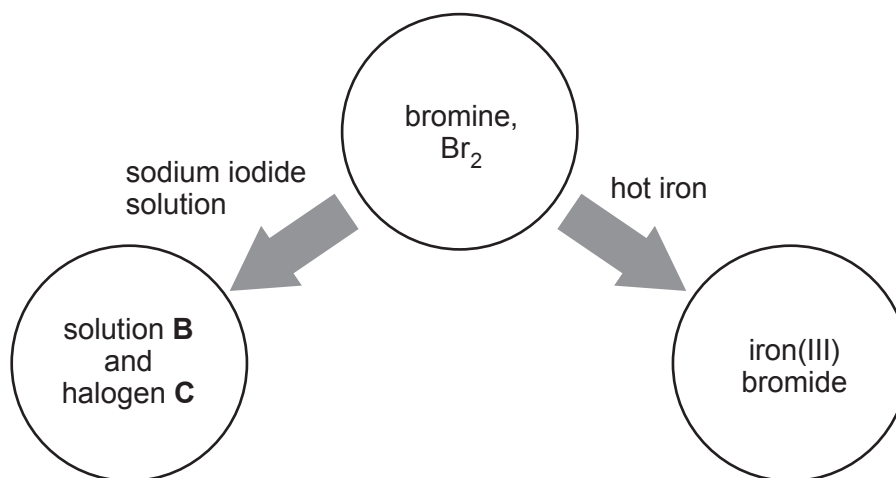


8. The diagram shows reactions of bromine,  $\text{Br}_2$ .



- (a) (i) Give the names of solution **B** and halogen **C**. [2]

solution **B** .....

halogen **C** .....

- (ii) Explain why bromine reacts with sodium iodide solution. [2]

.....

.....

.....

- (b) Write the symbol equation for the reaction between bromine,  $\text{Br}_2$ , and iron to form iron(III) bromide. [2]

.....

- (c) Silver nitrate solution was added to a solution of sodium iodide. Silver iodide was formed.

- (i) State what you would expect to see. [1]

.....

- (ii) Give the **ionic** equation for the formation of silver iodide. [2]

.....

END OF PAPER



### FORMULAE FOR SOME COMMON IONS

| POSITIVE IONS |                  | NEGATIVE IONS |                    |
|---------------|------------------|---------------|--------------------|
| Name          | Formula          | Name          | Formula            |
| aluminium     | $\text{Al}^{3+}$ | bromide       | $\text{Br}^-$      |
| ammonium      | $\text{NH}_4^+$  | carbonate     | $\text{CO}_3^{2-}$ |
| barium        | $\text{Ba}^{2+}$ | chloride      | $\text{Cl}^-$      |
| calcium       | $\text{Ca}^{2+}$ | fluoride      | $\text{F}^-$       |
| copper(II)    | $\text{Cu}^{2+}$ | hydroxide     | $\text{OH}^-$      |
| hydrogen      | $\text{H}^+$     | iodide        | $\text{I}^-$       |
| iron(II)      | $\text{Fe}^{2+}$ | nitrate       | $\text{NO}_3^-$    |
| iron(III)     | $\text{Fe}^{3+}$ | oxide         | $\text{O}^{2-}$    |
| lithium       | $\text{Li}^+$    | sulfate       | $\text{SO}_4^{2-}$ |
| magnesium     | $\text{Mg}^{2+}$ |               |                    |
| nickel        | $\text{Ni}^{2+}$ |               |                    |
| potassium     | $\text{K}^+$     |               |                    |
| silver        | $\text{Ag}^+$    |               |                    |
| sodium        | $\text{Na}^+$    |               |                    |
| zinc          | $\text{Zn}^{2+}$ |               |                    |





THE PERIODIC TABLE

Group

1 2 3 4 5 6 7 0

|                                                      |                                                      |                                                       |                                                      |                                                      |                                                       |                                                       |                                                       |                                                     |                                                       |                                                     |                                                     |                                                      |                                                      |                                                      |                                                       |                                                      |                                                    |                                                       |                                                  |
|------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------|--------------------------------------------------|
| <div><div>1<br/>H<br/>Hydrogen<br/>1</div></div>     |                                                      |                                                       |                                                      |                                                      |                                                       |                                                       |                                                       |                                                     |                                                       |                                                     |                                                     |                                                      |                                                      |                                                      |                                                       |                                                      |                                                    | <div><div>4<br/>He<br/>Helium<br/>2</div></div>       |                                                  |
| <div><div>7<br/>Li<br/>Lithium<br/>3</div></div>     | <div><div>9<br/>Be<br/>Beryllium<br/>4</div></div>   |                                                       |                                                      |                                                      |                                                       |                                                       |                                                       |                                                     |                                                       |                                                     |                                                     |                                                      |                                                      |                                                      |                                                       |                                                      |                                                    | <div><div>19<br/>F<br/>Fluorine<br/>9</div></div>     | <div><div>20<br/>Ne<br/>Neon<br/>10</div></div>  |
| <div><div>23<br/>Na<br/>Sodium<br/>11</div></div>    | <div><div>24<br/>Mg<br/>Magnesium<br/>12</div></div> |                                                       |                                                      |                                                      |                                                       |                                                       |                                                       |                                                     |                                                       |                                                     |                                                     |                                                      |                                                      |                                                      |                                                       |                                                      |                                                    | <div><div>35.5<br/>Cl<br/>Chlorine<br/>17</div></div> | <div><div>40<br/>Ar<br/>Argon<br/>18</div></div> |
| <div><div>39<br/>K<br/>Potassium<br/>19</div></div>  | <div><div>40<br/>Ca<br/>Calcium<br/>20</div></div>   | <div><div>45<br/>Sc<br/>Scandium<br/>21</div></div>   | <div><div>48<br/>Ti<br/>Titanium<br/>22</div></div>  | <div><div>51<br/>V<br/>Vanadium<br/>23</div></div>   | <div><div>52<br/>Cr<br/>Chromium<br/>24</div></div>   | <div><div>55<br/>Mn<br/>Manganese<br/>25</div></div>  | <div><div>56<br/>Fe<br/>Iron<br/>26</div></div>       | <div><div>59<br/>Co<br/>Cobalt<br/>27</div></div>   | <div><div>59<br/>Ni<br/>Nickel<br/>28</div></div>     | <div><div>63.5<br/>Cu<br/>Copper<br/>29</div></div> | <div><div>65<br/>Zn<br/>Zinc<br/>30</div></div>     | <div><div>70<br/>Ga<br/>Gallium<br/>31</div></div>   | <div><div>73<br/>Ge<br/>Germanium<br/>32</div></div> | <div><div>75<br/>As<br/>Arsenic<br/>33</div></div>   | <div><div>79<br/>Se<br/>Selenium<br/>34</div></div>   | <div><div>80<br/>Br<br/>Bromine<br/>35</div></div>   | <div><div>84<br/>Kr<br/>Krypton<br/>36</div></div> |                                                       |                                                  |
| <div><div>86<br/>Rb<br/>Rubidium<br/>37</div></div>  | <div><div>88<br/>Sr<br/>Strontium<br/>38</div></div> | <div><div>89<br/>Y<br/>Yttrium<br/>39</div></div>     | <div><div>91<br/>Zr<br/>Zirconium<br/>40</div></div> | <div><div>93<br/>Nb<br/>Niobium<br/>41</div></div>   | <div><div>96<br/>Mo<br/>Molybdenum<br/>42</div></div> | <div><div>99<br/>Tc<br/>Technetium<br/>43</div></div> | <div><div>101<br/>Ru<br/>Ruthenium<br/>44</div></div> | <div><div>103<br/>Rh<br/>Rhodium<br/>45</div></div> | <div><div>106<br/>Pd<br/>Palladium<br/>46</div></div> | <div><div>108<br/>Ag<br/>Silver<br/>47</div></div>  | <div><div>112<br/>Cd<br/>Cadmium<br/>48</div></div> | <div><div>115<br/>In<br/>Indium<br/>49</div></div>   | <div><div>119<br/>Sn<br/>Tin<br/>50</div></div>      | <div><div>122<br/>Sb<br/>Antimony<br/>51</div></div> | <div><div>128<br/>Te<br/>Tellurium<br/>52</div></div> | <div><div>127<br/>I<br/>Iodine<br/>53</div></div>    | <div><div>131<br/>Xe<br/>Xenon<br/>54</div></div>  |                                                       |                                                  |
| <div><div>133<br/>Cs<br/>Caesium<br/>55</div></div>  | <div><div>137<br/>Ba<br/>Barium<br/>56</div></div>   | <div><div>139<br/>La<br/>Lanthanum<br/>57</div></div> | <div><div>179<br/>Hf<br/>Hafnium<br/>72</div></div>  | <div><div>181<br/>Ta<br/>Tantalum<br/>73</div></div> | <div><div>184<br/>W<br/>Tungsten<br/>74</div></div>   | <div><div>186<br/>Re<br/>Rhenium<br/>75</div></div>   | <div><div>190<br/>Os<br/>Osmium<br/>76</div></div>    | <div><div>192<br/>Ir<br/>Iridium<br/>77</div></div> | <div><div>195<br/>Pt<br/>Platinum<br/>78</div></div>  | <div><div>197<br/>Au<br/>Gold<br/>79</div></div>    | <div><div>201<br/>Hg<br/>Mercury<br/>80</div></div> | <div><div>204<br/>Tl<br/>Thallium<br/>81</div></div> | <div><div>207<br/>Pb<br/>Lead<br/>82</div></div>     | <div><div>209<br/>Bi<br/>Bismuth<br/>83</div></div>  | <div><div>210<br/>Po<br/>Polonium<br/>84</div></div>  | <div><div>210<br/>At<br/>Astatine<br/>85</div></div> | <div><div>222<br/>Rn<br/>Radon<br/>86</div></div>  |                                                       |                                                  |
| <div><div>223<br/>Fr<br/>Francium<br/>87</div></div> | <div><div>226<br/>Ra<br/>Radium<br/>88</div></div>   | <div><div>227<br/>Ac<br/>Actinium<br/>89</div></div>  |                                                      |                                                      |                                                       |                                                       |                                                       |                                                     |                                                       |                                                     |                                                     |                                                      |                                                      |                                                      |                                                       |                                                      |                                                    |                                                       |                                                  |

Key

A<sub>r</sub>

Symbol

Name

Z

relative atomic mass

atomic number